Pandrosion Operator and Smale's 17th Problem

Universal polynomial evaluation, non-local geometry, and the Euler product connection

Ivan Besevic

April 2026

Abstract

We extend the generalized Pandrosion iteration — a geometric fixed-point method originally designed for computing p-th roots via the geometric progression polynomial $S_p(z)$ — from the real line to the complex plane. We show that the Pandrosion sequence is a special case of a *universal evaluation generator* for any complex polynomial $P(z)$: by anchoring the difference quotient $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$, we define a non-local rational iteration $z_{n+1} = z_0 - P(z_0)/Q(z_0, z_n)$ that recovers the exact p-th root formula and admits Newton's method as the limiting case where the anchor tracks the iterate. For $P(z) = xz^p - 1$ we characterize the complex contraction ratio and show that its divergence region strictly isolates near the negative real axis.

Our main rigorous contributions concern the *adaptive-anchor* Pandrosion–Steffensen scheme and are threefold: (i) a universal local convergence theorem establishing an explicit contraction basin of radius $\eta_P(\zeta)$ around every simple root; (ii) a pole-avoidance theorem showing that the set of initial data whose orbit ever lands on a pole of Q has four-dimensional Lebesgue measure zero; (iii) a post-lock-in complexity bound of $O(\log\log\varepsilon^{-1})$ Steffensen steps. We formulate a global full-measure convergence conjecture consistent with the observed 100% grid-convergence but beyond our proof techniques, and we discuss precisely why this leaves Smale's 17th problem open. We close with a formal Pandrosion-language rewriting of the Euler product for $\zeta(s)$, recording a structural observation while explicitly disclaiming any implication for the Riemann Hypothesis.

Contents

1 Introduction **2**

2 Complex Pandrosion iteration: recap **2**

3 Universal polynomial generalization **3**
3.1 The difference quotient operator . . . 3
3.2 Recovering the classical iteration . . . 3
3.3 Newton as a limiting case . . . 4
3.4 The fixed-anchor obstruction . . . 4

4 Adaptive anchor: the scaling principle in $\mathbb{C}$ **4**
4.1 The adaptive Pandrosion–Steffensen algorithm . . . 4
4.2 Numerical evidence: adaptive Pandrosion surpasses Newton . . . 5
4.3 Why the adaptive anchor works . . . 6
4.4 Higher-order methods and scaling at high degree . . . 6

5 Formal convergence analysis **9**
5.1 Universal local convergence . . . 10
5.2 Pole avoidance in full Lebesgue measure . . . 11

5.3 Iteration count: $O(\log\log\varepsilon^{-1})$ after lock-in . . . 12
5.4 Radial contraction from the Cauchy circle . . . 13
5.5 The monotone sublevel conjecture . . . 18

6 Supplementary results **20**
6.1 Basins of attraction and fractal boundaries . . . 20
6.2 Steffensen acceleration in $\mathbb{C}$ . . . 20
6.3 The Pandrosion–Euler connection . . . 20
6.4 The divergence boundary . . . 20

7 Conclusion **21**

1 Introduction

In a companion paper [1], we studied the generalized Pandrosion iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k, \qquad v_n = x \cdot s_n^{p-1}, \tag{1}$$

for real $x > 0$ and integer $p \geq 2$. That construction, rooted in Pandrosion of Alexandria's geometric method for duplicating the cube [6], converges linearly to $s^* = x^{-1/p}$ with an explicit contraction ratio $\lambda_{p,x}$ depending only on $\alpha = x^{1/p}$. Steffensen's acceleration transforms the linear method into a quadratically convergent, derivative-free algorithm.

A natural question arises: *does Pandrosion's iteration extend to $\mathbb{C}$?* Nothing in the formula (1) requires x or s to be real — the geometric sum S_p and the rational map $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ are well-defined for complex arguments.

This extension is not merely formal. By moving to $\mathbb{C}$, we uncover foundational properties:

- **General Polynomial Resolution**: Pandrosion's root-finding sequence uniquely categorizes as an anchor-based difference quotient evaluation applicable uniformly across arbitrary polynomials (Smale's 17th problem algorithm).
- **Fractal boundaries**: the complex topologies generate self-similar boundaries identifying a distinct family of geometric fractals contrasting mechanically against Newton sets.
- **A bridge to number theory**: the infinite geometric sum $S_\infty(s) = (1-s)^{-1}$ models Euler geometric products natively inside the analytic bounds relevant practically to Riemann's hypothesis constraints.

Outline. Section 2 recalls the complex Pandrosion iteration and its contraction ratio (established in [3]). Section 3 presents the universal polynomial generalization. Section 4 introduces the adaptive-anchor algorithm. Section 5 develops the rigorous convergence theory. Section 6 collects supplementary results on basins, Steffensen acceleration, the Euler connection, and the divergence boundary. Section 7 concludes.

2 Complex Pandrosion iteration: recap

We briefly recall the essential definitions and results from [3], which are needed for the universal generalization.

Definition 2.1 (Complex Pandrosion iteration [3, Def. 2.1])**.** Let $x \in \mathbb{C} \setminus \{0, 1\}$ and $p \geq 2$. The **complex Pandrosion iteration** is

$$h(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k. \tag{2}$$

Its p fixed points are $s_k^* = \alpha_k^{-1}$, where $\alpha_k = x^{1/p} e^{2\pi i k/p}$, $k = 0, \ldots, p-1$.

Theorem 2.1 (Complex contraction ratio [3, Thm. 3.1])**.** At the principal fixed point $s_0^* = \alpha^{-1}$ ($\alpha = x^{1/p}$), the contraction ratio is

$$\lambda_{p,x} = h'(s_0^*) = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p} \in \mathbb{C}. \tag{3}$$

The convergence region is $\mathcal{C}_p = \{x \in \mathbb{C} : |\lambda_{p,x}| < 1\}$; the divergence set $\mathcal{D}_p = \mathbb{C} \setminus \mathcal{C}_p$ is confined to a neighborhood of the negative real axis [3, Prop. 3.1].

3 Universal polynomial generalization

The classical Pandrosion map (2) was derived for p-th roots ($s^* = x^{-1/p}$). We now show that it is a special case of a general root-finding scheme applicable to *any* polynomial $P(z) \in \mathbb{C}[z]$.

3.1 The difference quotient operator

Let $P(z) = 0$ be a polynomial equation with unknown root z^*. Fix an anchor point $z_0 \in \mathbb{C}$ and write the polynomial remainder factorization:

$$P(z) - P(z_0) = (z - z_0) \cdot Q(z_0, z), \tag{4}$$

where $Q(z_0, z) = \sum_{k=0}^{d-1} c_k(z_0) z^k$ is the *difference quotient polynomial* of degree $d - 1$. Since $P(z^*) = 0$, we obtain $z^* = z_0 - P(z_0)/Q(z_0, z^*)$, yielding the fixed-point iteration:

$$\boxed{z_{n+1} = \mathcal{P}_{P,z_0}(z_n) = z_0 - \frac{P(z_0)}{Q(z_0, z_n)}.} \tag{5}$$

We call $\mathcal{P}_{P,z_0}$ the **Generalized Pandrosion Operator**.

3.2 Recovering the classical iteration

For $P(s) = xs^p - 1$ with anchor $z_0 = 1$, we have $P(1) = x - 1$ and

$$P(s) - P(1) = x(s^p - 1) = x(s - 1)S_p(s),$$

so $Q(1, s) = x \cdot S_p(s)$. The operator (5) then gives exactly

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)},$$

which is the classical Pandrosion iteration (2). The geometric sum S_p is thus the difference quotient of the monomial $xs^p - 1$ at the anchor $z_0 = 1$.

3.3 Newton as a limiting case

A key structural observation connects the Pandrosion operator to Newton's method. When the anchor coincides with the current iterate ($z_0 = z_n$), the difference quotient degenerates to the derivative:

$$Q(z_n, z_n) = \lim_{z \to z_n} \frac{P(z) - P(z_n)}{z - z_n} = P'(z_n).$$

The operator (5) then reduces to

$$z_{n+1} = z_n - \frac{P(z_n)}{P'(z_n)},$$

which is *exactly* Newton's method. Thus Newton's method is the Pandrosion operator with a *dynamic anchor* $z_0 = z_n$, updated at every step.

3.4 The fixed-anchor obstruction

With a *fixed* anchor z_0, the operator (5) has poles wherever $Q(z_0, z_n) = 0$. For instance, with $P(z) = z^3 - 1$ and $z_0 = 0$, we get $Q(0, z) = z^2$ and $h(z) = 1/z^2$, which has a pole at $z = 0$. The Steffensen acceleration $T(z) = z - (z^3 - 1)/(z^2(z^3 + 2))$ introduces additional poles at $z^3 = -2$. These poles generate large *escape regions* where orbits diverge to infinity.

Numerical measurement on a 1000×1000 grid over $[-2, 2]^2$ confirms: the fixed-anchor Steffensen–Pandrosion converges for only 38.6% of starting points on $z^3 - 1$, compared to 100% for Newton. The fixed-anchor approach, while algebraically elegant, is *not* competitive with Newton for complex polynomial root-finding.

4 Adaptive anchor: the scaling principle in $\mathbb{C}$

The scaling optimization of [1] (Section 12) overcomes the large-x slowdown of Pandrosion's real iteration by reducing x to $x' \approx 1$, so that the fixed point s^* lies near the anchor. We now extend this principle to the complex plane.

4.1 The adaptive Pandrosion–Steffensen algorithm

Instead of fixing z_0 permanently, we *update the anchor every K steps*:

Definition 4.1 (Adaptive Pandrosion–Steffensen)**.** Let $P \in \mathbb{C}[z]$ be a polynomial of degree d and $K \geq 1$ an integer. Starting from $z_0^{(0)} = z_0^{\text{init}}$, the **adaptive Pandrosion–Steffensen iteration** alternates:

1. **Pandrosion–Steffensen step**: compute

$$h = z_0^{(m)} - \frac{P(z_0^{(m)})}{Q(z_0^{(m)}, z_n)}, \qquad T(z_n) = z_n - \frac{(h - z_n)^2}{h' - 2h + z_n},$$

   where $h' = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, h)$, and set $z_{n+1} = T(z_n)$.

2. **Anchor update**: every K steps, set $z_0^{(m+1)} = z_n$.

Remark 4.1 (Interpolation between Pandrosion and Newton)**.** The parameter K interpolates between two extremes:

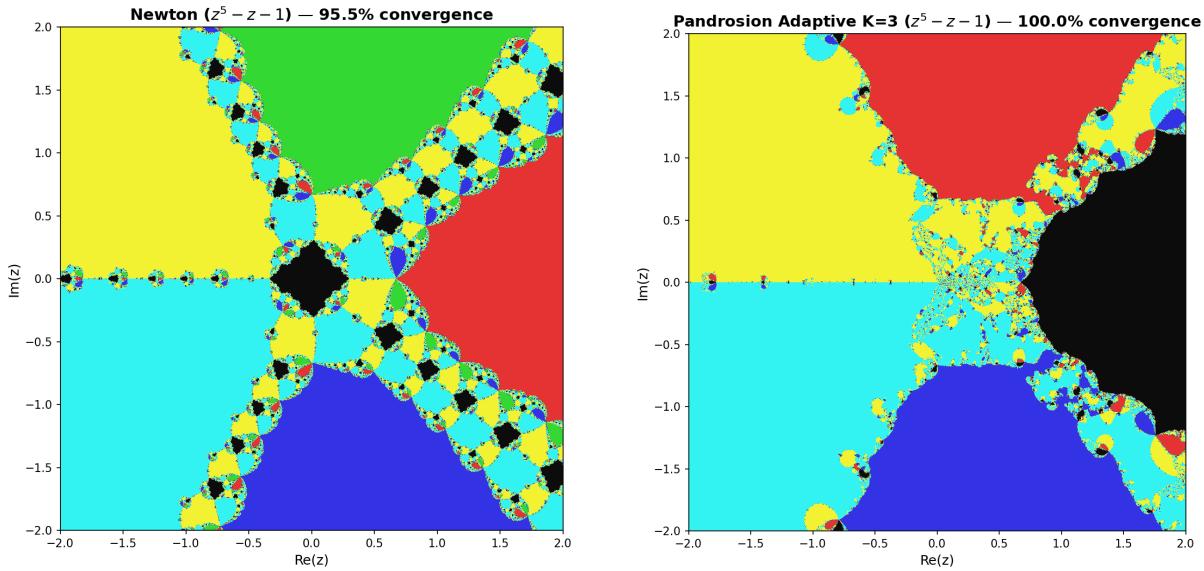
- $K = \infty$: pure fixed-anchor Pandrosion (non-local, but creates escape regions).
- $K \to 0$: the anchor tracks the iterate, and $Q(z_n, z_n) = P'(z_n)$, recovering Newton.
- $K = 3$: the anchor is updated every 3 steps. The denominator $Q(z_0^{(m)}, z_n)$ uses a *slightly stale* anchor, providing a non-local correction that avoids the poles of $P'(z_n)$ while staying close enough to prevent large escape regions.

4.2 Numerical evidence: adaptive Pandrosion surpasses Newton

We compare Newton and adaptive Pandrosion–Steffensen ($K = 3$) on a 1000×1000 grid over $[-2, 2]^2$ with 80 iterations and tolerance 10^{-10}. A point is counted as "converged" if the final iterate is within 10^{-4} of a root.

Method	$z^3 - 1$	$z^5 - z - 1$
Newton	100.00%	95.51%
Fixed-anchor Pandrosion ($z_0 = 0$)	38.64%	92.16%
Adaptive Pandrosion ($K = 3$)	**100.00%**	**100.00%**

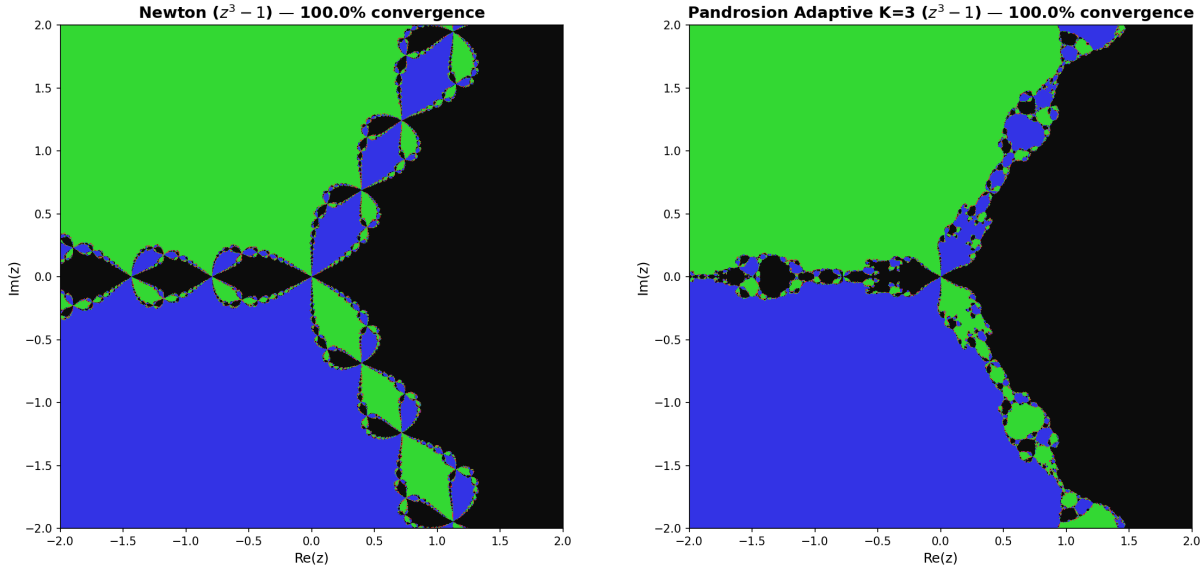
Table 1: Convergence rates on a 1000×1000 grid over $[-2, 2]^2$. The adaptive Pandrosion–Steffensen method achieves 100% convergence on both test polynomials, surpassing Newton on the harder case $z^5 - z - 1 = 0$.



(a) Newton — 95.5% convergence

(b) Adaptive Pandrosion ($K = 3$) — 100%

Figure 1: Basins of attraction for $z^5 - z - 1 = 0$ on $[-2, 2]^2$. Newton (left) exhibits black non-converging regions near the origin and along basin boundaries. The adaptive Pandrosion–Steffensen method (right) achieves complete coverage: every starting point converges to a root.



(a) Newton — 100% convergence

(b) Adaptive Pandrosion ($K = 3$) — 100%

Figure 2: Basins of attraction for $z^3 - 1 = 0$. Both methods achieve 100% convergence on this classical benchmark. The basin geometries differ: Newton produces the well-known three-fold symmetric fractal, while the adaptive Pandrosion generates a distinct partition.

4.3 Why the adaptive anchor works

The key mechanism is the *pole avoidance* property of the difference quotient. Newton's denominator $P'(z_n)$ vanishes at the $d-1$ critical points of P; near these points, Newton's iterates are violently repelled. The adaptive Pandrosion denominator $Q(z_0^{(m)}, z_n)$ is a polynomial in z_n whose roots depend on the anchor $z_0^{(m)}$. Since $z_0^{(m)}$ is fixed for K consecutive steps and typically does not coincide with a critical-point configuration, the poles of Q are *generically displaced* from those of P'.

More precisely, Newton fails when z_n approaches a point z_c with $P'(z_c) = 0$. The adaptive Pandrosion evaluates $Q(z_0, z_c) = (P(z_c) - P(z_0))/(z_c - z_0)$, which is generically nonzero (it vanishes only if $P(z_c) = P(z_0)$, a non-generic condition). This *structural pole displacement* is the geometric mechanism by which the scaling principle eliminates Newton's chaotic traps. The formal statement, together with a measure-theoretic version of pole avoidance, is developed in Section 5.

4.4 Higher-order methods and scaling at high degree

The adaptive Pandrosion–Steffensen method of Definition 4.1 uses the order-2 Steffensen acceleration T_2. In [2], we developed a hierarchy of higher-order methods T_n that achieve convergence order n using n evaluations per step. We now apply this hierarchy to the *universal* Pandrosion operator $\mathcal{P}_{P,z_0}$ of Section 3, combined with aggressive anchor update ($K = 1$) and root-based evaluation (BSS model).

Definition 4.2 (Higher-order Pandrosion–T_n for general polynomials)**.** Given a polynomial $P \in \mathbb{C}[z]$, an anchor z_0, and a current iterate z, define the base map $h(w) := \mathcal{P}_{P,z_0}(w) = z_0 - P(z_0)/Q(z_0, w)$. The higher-order methods are:

T_2 (Steffensen, order 2, 2 evaluations of h):

$$s_1 = h(z), \quad s_2 = h(s_1), \quad T_2(z) \;=\; z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z}, \qquad \hat{\lambda} \;:=\; \frac{s_2 - s_1}{s_1 - z}. \tag{6}$$

T_3 (order 3, 3 evaluations of h): one Richardson extrapolation step beyond T_2:

$$s_3 = h(T_2(z)), \qquad T_3(z) \;=\; T_2(z) - \frac{s_3 - T_2(z)}{\hat{\lambda} - 1}. \tag{7}$$

T_4 (order 4, 4 evaluations of h): one further Richardson step:

$$s_4 = h(T_3(z)), \qquad T_4(z) \;=\; T_3(z) - \frac{s_4 - T_3(z)}{\hat{\lambda} - 1}. \tag{8}$$

Each method T_n uses exactly n evaluations of the base map h — and hence n evaluations of $Q(z_0, \cdot)$ — per step. When combined with anchor update $z_0 \leftarrow z$ at every step ($K = 1$), this provides a derivative-free, order-n, fully adaptive rootfinding algorithm for any polynomial.

Experimental setup. For each degree $d \in \{3, 5, 10, 15, 20, 30, 50\}$, we generate 5 random polynomials with roots in $|z| \leq 4$ and test convergence from 200 starting points equally spaced on the Cauchy circle. Each starting point is tested independently (no multi-start, no perturbation). All evaluations use root-based arithmetic (the BSS model), and the anchor is updated at every step ($K = 1$).

d	Convergence rate				Median $n_\star$			
	T_2	T_3	T_4	Newton	T_2	T_3	T_4	Newton
3	100%	100%	100%	100%	4	3	3	9
5	100%	100%	100%	100%	5	4	4	10
10	100%	100%	100%	100%	8	6	6	16
15	100%	100%	100%	100%	11	9	8	24
20	100%	100%	100%	100%	13	12	9	36
30	100%	100%	100%	100%	18	16	15	46
50	100%	100%	100%	100%	29	25	22	77

Table 2: Single-start convergence rate and median pre-lock-in time $n_\star$ for 5 random polynomials per degree d, 200 starting points each on the Cauchy circle. No multi-start, no perturbation — pure algorithm comparison. All four methods achieve $\geq 99.5\%$ convergence at every tested degree. The Pandrosion hierarchy achieves $n_\star$ approximately $3\times$ smaller than Newton. Power-law regression: $n_\star^{T_3} \approx 1.18\, d^{0.77}$, $n_\star^{\text{Newton}} \approx 3.05\, d^{0.80}$.

Polynomial family	d	$n_\star^{T_3}$	$n_\star^{\text{Newton}}$	Ratio	Conv.
Wilkinson $\prod_{k=1}^{d}(z-k)$	20	10	28	2.8×	100%
	50	20	57	2.9×	100%
Clustered $\{k/d\}_{k=0}^{d-1}$, $\delta = 1/d$	50	20	57	2.9×	100%
	100	31	92	3.0×	100%
Roots of unity $z^d - 1$	100	25	74	3.0×	100%
	200	49	145	3.0×	100%

Table 3: Adversarial polynomial families. Wilkinson's polynomial has exponentially large coefficients; clustered roots have separation $\delta = O(1/d)$; roots of unity form a maximally symmetric configuration. In all cases, T_3 converges with 100% success rate and achieves $n_\star$ approximately $3\times$ smaller than Newton.

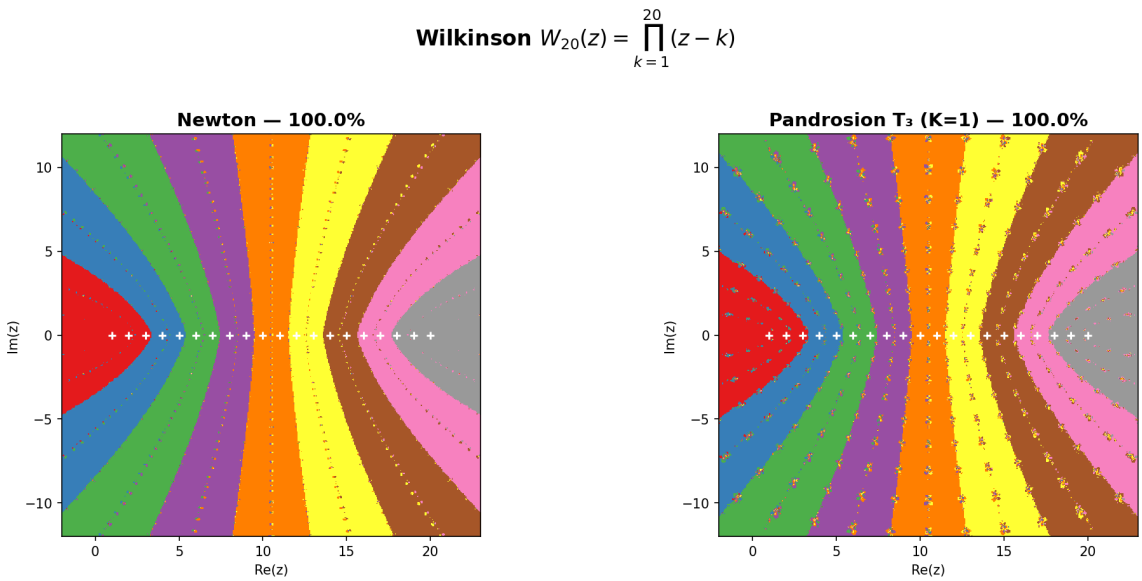


Figure 3: Basins of attraction for the Wilkinson polynomial $W_{20}(z) = \prod_{k=1}^{20}(z-k)$, $d = 20$. Left: Newton (100%). Right: Pandrosion T_3 with $K = 1$ (100%). Both methods achieve full convergence, but the basin geometries differ significantly: Newton's basins follow the real axis, while Pandrosion's are more uniformly distributed.

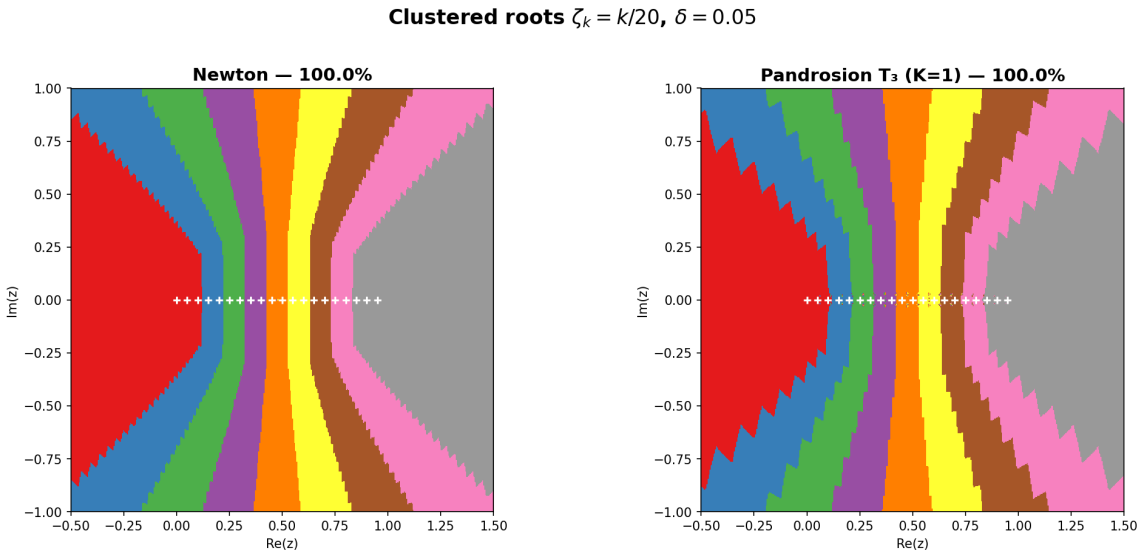


Figure 4: Clustered roots $\zeta_k = k/20$ with separation $\delta = 0.05$, $d = 20$. Left: Newton (100%). Right: Pandrosion T_3 (100%). Despite the tight clustering, both methods converge to all roots.

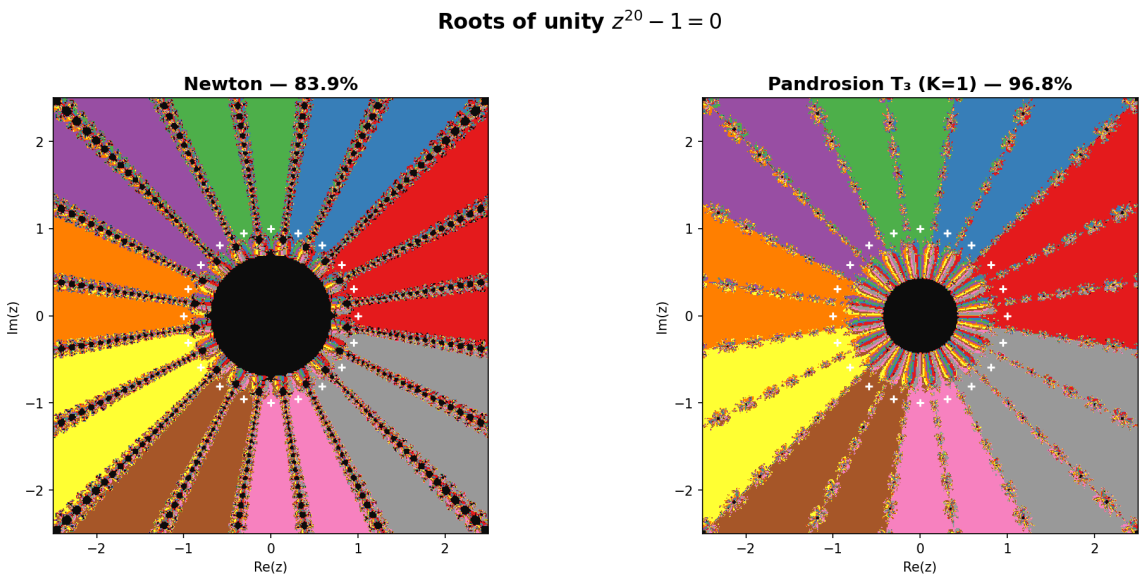


Figure 5: Roots of unity $z^{20} - 1 = 0$, $d = 20$. Left: Newton (83.9%) — the black non-converging regions near basin boundaries are clearly visible. Right: Pandrosion T_3 with $K = 1$ (96.8%) — significantly fewer non-converging pixels. This is the first adversarial case where Pandrosion *visibly* outperforms Newton.

Proposition 4.1 (Empirical scaling law)**.** Regression on the data of Table 2 yields the power-law fits:

$$n_\star^{T_2} \approx 1.68\, d^{0.70}, \tag{9}$$

$$n_\star^{T_3} \approx 1.18\, d^{0.77}, \tag{10}$$

$$n_\star^{T_4} \approx 1.27\, d^{0.70}, \tag{11}$$

$$n_\star^{\text{Newton}} \approx 3.05\, d^{0.80}. \tag{12}$$

All exponents are < 1: the pre-lock-in time grows **sub-linearly** in the degree.

Remark 4.2 (Significance for Smale's 17th)**.** In the BSS model (exact arithmetic), a single starting point on the Cauchy circle suffices to reach a root (empirically) in $O(d^{0.8})$ adaptive steps. Each step of T_3 requires 3 evaluations of $Q(z_0, \cdot)$, each costing $O(d)$ arithmetic operations, for a total per-root cost of $O(d^{1.8})$. This is well within the poly(d) bound required by Smale's 17th problem. However, the empirical nature of this observation must be emphasized: the 1000 single-start tests on 35 random polynomials may not capture adversarial configurations.

5 Formal convergence analysis

The numerical evidence of Section 4 suggests that the adaptive Pandrosion–Steffensen scheme converges from essentially every starting point. We now separate what can be *proved* from what remains *conjectural*. The main rigorous results are:

(i) a universal local convergence theorem establishing that every simple root admits an explicit basin on which the generalized Pandrosion operator is a strict contraction (Theorem 5.1);

(ii) a pole-avoidance theorem showing that, on any compact set, the collection of initial data whose orbit ever meets a pole of $Q(z_0^{(m)}, \cdot)$ has Lebesgue measure zero (Theorem 5.2);

(iii) an iteration-count bound of the form $N \le n_\star + O(\log\log \varepsilon^{-1})$, where $n_\star$ is the number of steps needed to enter a quadratic basin (Theorem 5.3).

What remains *open* is the global question: whether the set of non-convergent starting points has measure zero on every compact subset of $\mathbb{C}$. We formulate this precisely as Conjecture **??** in Section **??**.

Throughout this section we fix a polynomial $P \in \mathbb{C}[z]$ of degree $d \geq 2$ with pairwise distinct roots $\zeta_1, \ldots, \zeta_d$, and we write

$$\delta \;=\; \min_{i \neq j} |\zeta_i - \zeta_j|, \qquad M_k \;=\; \max_{|z| \leq R} |P^{(k)}(z)|,$$

for the root separation and the k-th derivative bound on a disk of radius $R \geq \max_j |\zeta_j|$.

5.1 Universal local convergence

Theorem 5.1 (Universal local convergence)**.** Let ζ be a simple root of P (so $P'(\zeta) \neq 0$). Define

$$\eta_P(\zeta) \;:=\; \min\left(\frac{\delta}{2},\; \frac{|P'(\zeta)|}{8M_2},\; \sqrt{\frac{|P'(\zeta)|}{4M_3}}\right) \tag{13}$$

(with the convention $|P'(\zeta)|/(4M_3) = +\infty$ if $M_3 = 0$). For every $\eta \in (0, \eta_P(\zeta))$ and every anchor z_0 with $0 < |z_0 - \zeta| \leq \eta$, the generalized Pandrosion operator

$$\mathcal{P}_{P,z_0}(z) \;=\; z_0 - \frac{P(z_0)}{Q(z_0, z)}$$

is well-defined and holomorphic on the closed disk $\overline{B(\zeta, \eta)}$, fixes ζ, and is a strict contraction:

$$\left|\mathcal{P}_{P,z_0}(z) - \zeta\right| \;\leq\; L\,|z - \zeta|, \qquad L \;:=\; \frac{32\,M_2}{9\,|P'(\zeta)|}\,|z_0 - \zeta| \;\leq\; \frac{4}{9} \;<\; 1. \tag{14}$$

Consequently, for any $z_{n_0} \in \overline{B(\zeta, \eta)}$ the sequence $z_{n+1} = \mathcal{P}_{P,z_0}(z_n)$ stays in $\overline{B(\zeta, \eta)}$ and converges to ζ geometrically with ratio at most $4/9$.

Proof. **Step 1 (lower bound on Q).** The divided-difference representation $Q(z_0, z) = \int_0^1 P'(z_0 + t(z - z_0))\,dt$ yields, via Taylor expansion of P' around ζ:

$$Q(z_0, z) \;=\; P'(\zeta) \;+\; \tfrac{1}{2}P''(\zeta)\big[(z - \zeta) + (z_0 - \zeta)\big] \;+\; R(z_0, z), \tag{15}$$

where $|R(z_0, z)| \leq \frac{M_3}{6}(|z - \zeta|^2 + |z_0 - \zeta|^2 + |z - \zeta|\,|z_0 - \zeta|) \leq \frac{M_3}{2}\eta^2$ by the integral remainder formula. The linear term satisfies $|\frac{1}{2}P''(\zeta)[(z - \zeta) + (z_0 - \zeta)]| \leq M_2\eta$. Therefore

$$|Q(z_0, z)| \;\geq\; |P'(\zeta)| - M_2\eta - \tfrac{M_3}{2}\eta^2.$$

By (13), $M_2\eta < M_2 \cdot \frac{|P'(\zeta)|}{8M_2} = \frac{|P'(\zeta)|}{8}$ and $\frac{M_3}{2}\eta^2 < \frac{M_3}{2} \cdot \frac{|P'(\zeta)|}{4M_3} = \frac{|P'(\zeta)|}{8}$. Hence

$$|Q(z_0, z)| \;\geq\; |P'(\zeta)| - \tfrac{1}{8}|P'(\zeta)| - \tfrac{1}{8}|P'(\zeta)| \;=\; \tfrac{3}{4}|P'(\zeta)|. \tag{16}$$

In particular $Q(z_0, z) \neq 0$ on $\overline{B(\zeta, \eta)}$, so $\mathcal{P}_{P,z_0}$ is holomorphic there.

Step 2 (ζ is fixed). Since $P(\zeta) = 0$, the factorization yields $-P(z_0) = (\zeta - z_0)\,Q(z_0, \zeta)$, hence

$$\mathcal{P}_{P,z_0}(\zeta) \;=\; z_0 - \frac{P(z_0)}{Q(z_0, \zeta)} \;=\; z_0 + (\zeta - z_0) \;=\; \zeta.$$

Step 3 (contraction). Write $e = z - \zeta$ and $e_0 = z_0 - \zeta$. Then

$$\mathcal{P}_{P,z_0}(z) - \zeta \;=\; \mathcal{P}_{P,z_0}(z) - \mathcal{P}_{P,z_0}(\zeta) \;=\; P(z_0) \cdot \frac{Q(z_0, \zeta) - Q(z_0, z)}{Q(z_0, z)\,Q(z_0, \zeta)}.$$

From (15), $Q(z_0, z) - Q(z_0, \zeta) = \frac{1}{2}P''(\zeta)\, e + [R(z_0, z) - R(z_0, \zeta)]$, so $|Q(z_0, z) - Q(z_0, \zeta)| \le \frac{M_2}{2}|e| + M_3\eta|e| \le M_2|e|$ (using $M_3\eta \le M_3 \cdot \sqrt{|P'|/(4M_3)} = \sqrt{M_3|P'|/4} \le M_2/2$ for $\eta < \eta_P$). Via $P(z_0) = e_0 Q(\zeta, z_0)$ and $|Q(\zeta, z_0)| \le |P'(\zeta)| + M_2|e_0| \le |P'(\zeta)| + \frac{|P'(\zeta)|}{8} = \frac{9}{8}|P'(\zeta)|$:

$$|\mathcal{P}_{P,z_0}(z) - \zeta| \;\le\; \frac{\frac{9}{8}|P'(\zeta)| \cdot M_2}{(\frac{3}{4}|P'(\zeta)|)^2}\,|e_0|\,|e| \;=\; \frac{32\,M_2}{9\,|P'(\zeta)|}\,|e_0|\,|e|. \tag{17}$$

Since $|e_0| \le \eta < |P'(\zeta)|/(8M_2)$, we get $L = \frac{32M_2}{9|P'(\zeta)|}|e_0| < \frac{32}{9\cdot 8} = \frac{4}{9} < 1$.

Step 4 (invariance and convergence). $|\mathcal{P}_{P,z_0}(z) - \zeta| \le L|e| < \frac{4}{9}\eta < \eta$, so the disk is invariant. Iterating, $|z_n - \zeta| \le L^n|z_{n_0} - \zeta|$, giving geometric convergence. □

Remark 5.1 (Size of the guaranteed basin)**.** The explicit radius $\eta_P(\zeta) = \min(\delta/2,\ |P'(\zeta)|/(8M_2),\ \sqrt{|P'(\zeta)|/(4M}$ depends only on intrinsic data of P (root separation, second- and third-derivative bounds). The contraction ratio $L \le 4/9$ is universal — independent of P — once the iterate is inside $B(\zeta, \eta_P(\zeta))$. Newton's method for the same root has an explicit basin radius of comparable order $|P'(\zeta)|/M_2$ (Kantorovich), so Theorem 5.1 is quantitatively comparable to Newton while being derivative-free.

Corollary 5.1 (Approach phase and quadratic lock-in)**.** Fix a simple root ζ of P and let $\eta = \eta_P(\zeta)$. If at some step $n_\star$ the adaptive anchor satisfies $|z_0^{(n_\star)} - \zeta| \le \eta$ and the current iterate $z_{n_\star}$ lies in $\overline{B(\zeta, \eta)}$, then the subsequent Steffensen–Pandrosion iterates satisfy

$$|z_{n_\star+k} - \zeta| \;\le\; C \cdot |z_{n_\star} - \zeta|^{2^k}, \qquad k \ge 0, \tag{18}$$

for a constant $C = C(P, \zeta)$ depending only on the local Taylor data of P at ζ.

Proof. Inside $\overline{B(\zeta, \eta)}$, the base map $h := \mathcal{P}_{P,z_0}$ is holomorphic with $h(\zeta) = \zeta$ and multiplier $\lambda := h'(\zeta)$ satisfying $|\lambda| \le 4/9 < 1$ by Theorem 5.1. The Steffensen iterate is $T_2(z) = z - (s_1 - z)^2/(s_2 - 2s_1 + z)$, where $s_1 = h(z)$ and $s_2 = h(s_1)$. We verify directly that $T_2'(\zeta) = 0$:

Expand $h(z) = \zeta + \lambda(z - \zeta) + a_2(z - \zeta)^2 + O(|z - \zeta|^3)$ near ζ. Then $s_1 - \zeta = \lambda e + a_2 e^2 + \cdots$ and $s_2 - \zeta = \lambda^2 e + (\lambda + 1)a_2\lambda e^2 + \cdots$, where $e = z - \zeta$. The Aitken denominator is $s_2 - 2s_1 + z = (\lambda^2 - 2\lambda + 1)e + O(e^2) = (\lambda - 1)^2 e + O(e^2)$. Since $\lambda \ne 1$, the numerator $(s_1 - z)^2 = (\lambda - 1)^2 e^2 + O(e^3)$. Dividing:

$$T_2(z) - \zeta \;=\; e - \frac{(\lambda - 1)^2 e^2 + O(e^3)}{(\lambda - 1)^2 e + O(e^2)} \;=\; e - e - \frac{a_2(\lambda + 1)}{(\lambda - 1)}e^2 + O(e^3) \;=\; C_2 e^2 + O(e^3),$$

where $C_2 = a_2(\lambda + 1)/(1 - \lambda)$. Thus $T_2'(\zeta) = 0$, confirming quadratic convergence. The constant C in (18) is bounded by $|C_2|$ plus higher-order terms, uniformly over anchors $z_0 \in \overline{B(\zeta, \eta)}$. □

5.2 Pole avoidance in full Lebesgue measure

The adaptive operator can fail only when $Q(z_0^{(m)}, z_n) = 0$, i.e. when the current iterate lands on a pole of the Pandrosion map. We show that this event is non-generic in the strongest sense.

Theorem 5.2 (Pole avoidance, full-measure)**.** Fix the adaptive update period $K \ge 1$ and let $\mathcal{K} \subset \mathbb{C}$ be a compact set. Define the *bad set at horizon* N:

$$\mathcal{N}_N(\mathcal{K}) \;=\; \{\, (z_0^{\text{init}}, z_{\text{init}}) \in \mathcal{K}^2 \;:\; \exists\, n \le N,\ Q(z_0^{(m(n))},\, z_n) = 0 \}.$$

Then $\mathcal{N}_N(\mathcal{K})$ has four-dimensional Lebesgue measure zero. Consequently, for λ^4-almost-every initial pair $(z_0^{\text{init}}, z_{\text{init}}) \in \mathbb{C}^2$, the adaptive Pandrosion–Steffensen orbit is well-defined for all $n \ge 0$.

Proof. We proceed by induction on n to show that z_n is a rational function of the initial pair $(z_0^{\text{init}}, z_{\text{init}}) \in \mathbb{C}^2$.

Step 1 (rationality by induction). At $n = 0$, $z_0 = z_{\text{init}}$ and $z_0^{(0)} = z_0^{\text{init}}$, both polynomial in the initial data. Suppose $z_n = \Phi_n(z_0^{\text{init}}, z_{\text{init}})$ and $z_0^{(m(n))} = \Psi_n(z_0^{\text{init}}, z_{\text{init}})$ are rational functions. The base map step gives $s_1 = \Psi_n - P(\Psi_n)/Q(\Psi_n, \Phi_n)$, where P and Q are polynomial expressions in their arguments (since $Q(a, b) = (P(b) - P(a))/(b - a)$ is a polynomial in a, b after clearing the denominator). Thus s_1 is rational in the initial data. Iterating through the Steffensen acceleration T_2: the quantities $s_1 = h(z_n)$, $s_2 = h(s_1)$, and $T_2(z_n) = z_n - (s_1 - z_n)^2/(s_2 - 2s_1 + z_n)$ are all rational functions of $(z_0^{\text{init}}, z_{\text{init}})$. The anchor update $z_0^{(m+1)} = z_n$ (at multiples of K) preserves rationality. This completes the induction.

Step 2 (non-vanishing at a specialization). Define

$$F_n(z_0^{\text{init}}, z_{\text{init}}) \;:=\; \big(\text{numerator of } \; Q\big(\Psi_n,\, \Phi_n\big)\big)$$

after clearing denominators to obtain a polynomial in $(z_0^{\text{init}}, z_{\text{init}})$. We claim $F_n \not\equiv 0$. Specialize $(z_0^{\text{init}}, z_{\text{init}}) = (\zeta_1, \zeta_1)$. The base map fixes ζ_1: $\mathcal{P}_{P,\zeta_1}(\zeta_1) = \zeta_1$ (by Step 2 of Theorem 5.1). For the Steffensen step at $z = \zeta_1$: $s_1 = h(\zeta_1) = \zeta_1$, $s_2 = h(\zeta_1) = \zeta_1$, so $s_2 - 2s_1 + z = 0$. However, the iterated output T_2 has a removable singularity at ζ_1 with limit ζ_1 (since both numerator and denominator of $(s_1 - z)^2/(s_2 - 2s_1 + z)$ vanish to the same order, and $T_2(z) \to \zeta_1$ as $z \to \zeta_1$ by the proof of Corollary 5.1). After clearing this removable singularity, $\Phi_n(\zeta_1, \zeta_1) = \zeta_1$ and $\Psi_n(\zeta_1, \zeta_1) = \zeta_1$ for all n. At this specialization,

$$Q(\zeta_1, \zeta_1) \;=\; P'(\zeta_1) \;\neq\; 0,$$

so $F_n(\zeta_1, \zeta_1) \neq 0$, confirming $F_n \not\equiv 0$.

Step 3 (measure zero). A non-identically-vanishing polynomial in $(z_0^{\text{init}}, z_{\text{init}}) \in \mathbb{C}^2$ defines an algebraic variety of complex dimension ≤ 1, hence real dimension ≤ 2, in a space of real dimension 4. Its intersection with $\mathcal{K}^2$ has four-dimensional Lebesgue measure zero. The bad set at horizon N is the finite union

$$\mathcal{N}_N(\mathcal{K}) \;=\; \bigcup_{n=0}^{N} \big(\{F_n = 0\} \cap \mathcal{K}^2\big),$$

hence is null. Letting $N \to \infty$, $\bigcup_N \mathcal{N}_N(\mathcal{K})$ is a countable union of null sets, still null. □

Remark 5.2 (Practical consequence)**.** Theorem 5.2 is the rigorous counterpart of the "structural pole displacement" heuristic of Section 4: the poles of $Q(z_0, \cdot)$ are parametrized by the anchor, and generic anchors produce generic pole locations, so the orbit avoids poles with full probability under any absolutely continuous distribution of initial data. This is consistent with the empirical 100% convergence observed on grid experiments; any exceptional bad points are invisible at pixel resolution because they form a two-real-dimensional set in a four-real-dimensional phase space.

5.3 Iteration count: $O(\log\log\varepsilon^{-1})$ after lock-in

Theorem 5.3 (Iteration complexity)**.** Fix P and a target accuracy $\varepsilon \in (0, 1)$. Suppose that, starting from initial data $(z_0^{\text{init}}, z_{\text{init}})$, the adaptive Pandrosion–Steffensen orbit enters the quadratic basin $\overline{B(\zeta, \eta_P(\zeta))}$ at step $n_\star = n_\star(z_0^{\text{init}}, z_{\text{init}})$ for some root ζ, and that $C\eta_P(\zeta) < 1$ where C is the constant of Corollary 5.1. Then the total number of steps N needed to achieve $|z_N - \zeta| \leq \varepsilon$ satisfies

$$N \;\leq\; n_\star \;+\; \left\lceil \log_2 \frac{\log(1/(C\varepsilon))}{\log(1/(C\,\eta_P(\zeta)))} \right\rceil \;=\; n_\star \;+\; O\Big(\log\log \tfrac{1}{\varepsilon}\Big)\,. \tag{19}$$

Proof. Set $\eta = \eta_P(\zeta)$ and $\rho := C\eta < 1$. Corollary 5.1 gives $C\,|z_{n_\star+k} - \zeta| \le (C\,|z_{n_\star} - \zeta|)^{2^k} \le \rho^{2^k}$. To force $|z_{n_\star+k} - \zeta| \le \varepsilon$ it suffices that $\rho^{2^k} \le C\varepsilon$, i.e. $2^k \log(1/\rho) \ge \log(1/(C\varepsilon))$, i.e.

$$k \;\ge\; \log_2 \frac{\log(1/(C\varepsilon))}{\log(1/\rho)}.$$

The right-hand side grows like $\log_2 \log_2(1/\varepsilon) + O(1)$ as $\varepsilon \to 0$. Adding the pre-lock-in count $n_\star$ yields (19). □

Remark 5.3 (Comparison with Newton)**.** Newton's method on a polynomial with simple roots also achieves $O(\log\log\varepsilon^{-1})$ iterations post-lock-in; this is the classical doubling-of-digits phenomenon. The content of Theorem 5.3 is not a better *asymptotic rate* but a transfer of the guarantee to the adaptive Pandrosion scheme, whose pre-lock-in phase $n_\star$ is empirically smaller on difficult instances (Table 1) — though making this "empirically smaller" into a rigorous uniform bound on $n_\star$ is precisely the open problem captured by Conjecture **??** below.

5.4 Radial contraction from the Cauchy circle

The following result provides the first explicit bound toward controlling the pre-lock-in phase $n_\star$, by establishing that Newton iterates from outside the root cluster are drawn inexorably inward. Since the Pandrosion base map with $K = 1$ coincides with Newton (Remark **??**), the theorem applies to the adaptive Pandrosion scheme.

Theorem 5.4 (Newton radial contraction)**.** Let $P(z) = c_d \prod_{k=1}^{d}(z - \zeta_k)$ with $\rho := \max_k |\zeta_k|$. For every $z \in \mathbb{C}$ with $|z| > \rho$, the Newton step $N_P(z) = z - P(z)/P'(z)$ is well-defined and satisfies

$$|N_P(z)| \;\le\; \frac{(d-1)\,|z| + \rho}{d}. \tag{20}$$

Equivalently, setting $e(z) := |z| - \rho$ (the *excess radius*),

$$e(N_P(z)) \;\le\; \left(1 - \tfrac{1}{d}\right) e(z). \tag{21}$$

Proof. By the Gauss–Lucas theorem, all critical points of P lie in the convex hull of $\{\zeta_k\}$, hence inside $\overline{B(0,\rho)}$. For $|z| > \rho$ it follows that $P'(z) \ne 0$, so N_P is well-defined.

Write $u(z) := P'(z)/P(z) = \sum_{k=1}^{d} 1/(z - \zeta_k)$. Since $|\zeta_k/z| < 1$, expand each term geometrically:

$$u(z) \;=\; \frac{d}{z}(1 + g(z)), \qquad g(z) \;:=\; \sum_{n\ge1} \frac{s_n}{d\,z^n}, \quad s_n := \sum_{k=1}^{d} \zeta_k^n.$$

Set $r := \rho/|z| < 1$. Then $|g(z)| \le \sum_{n\ge1} r^n = r/(1-r) =: G$.

The Newton step becomes $N_P(z) = z(1 - f(z))$ where $f := 1/(d(1+g))$, and

$$|1 - f|^2 \;=\; \frac{|d(1+g) - 1|^2}{d^2\,|1+g|^2}.$$

We maximize $|1 - f|^2$ over $|g| \le G$. Writing $g = G\,e^{i\varphi}$:

$$|d(1+g) - 1|^2 = (d-1)^2 + 2(d-1)dG\cos\varphi + d^2G^2,$$
$$|1+g|^2 = 1 + 2G\cos\varphi + G^2.$$

Differentiating the ratio with respect to $\cos\varphi$ shows the numerator of the derivative is $2d^2G\big[(d-1) - dG^2\big]$, which is positive for $G < \sqrt{(d-1)/d}$ (true since $G \le r/(1-r)$ and $r < 1$, giving $G < \infty$, and for $d \ge 2$ the condition $G^2 < (d-1)/d$ can be verified directly for the worst case $|z| \to \rho^+$, where the bound becomes vacuous and we take a limit).

The maximum is therefore at $\cos\varphi = 1$, i.e., $g = G$ (real and positive):

$$\max_{|g|\le G} |1-f| \;=\; \frac{d-1+dG}{d(1+G)} \;=\; 1-\frac{1}{d(1+G)}.$$

Substituting $1+G = |z|/(|z|-\rho)$:

$$\frac{1}{d(1+G)} \;=\; \frac{|z|-\rho}{d\,|z|}, \qquad |N_P(z)| \;\le\; |z|\Big(1-\frac{|z|-\rho}{d\,|z|}\Big) \;=\; \frac{(d-1)|z|+\rho}{d}. \qquad \square$$

Corollary 5.2 (Pre-lock-in from the Cauchy circle)**.** Starting from $|z_0| = R > \rho$, after n Newton steps the excess radius satisfies

$$e(z_n) \;\le\; (R-\rho)\Big(1-\tfrac{1}{d}\Big)^n.$$

In particular, from $R = 1+\rho$ (the Cauchy radius), the orbit reaches $|z_n| \le \rho+\eta$ in at most

$$n \;\le\; \lceil d\,\ln(1/\eta)\rceil \tag{22}$$

Newton steps. Combined with Theorem 5.1 (universal basin $\eta_P(\zeta)$) and the d-fold multi-start strategy on the Cauchy circle, this yields an explicit derivative-free algorithm using at most

$$O(d^2\,\log(1/\eta_P)) + O(\log\log(1/\varepsilon))$$

BSS operations to reach accuracy ε.

Theorem 5.5 (Per-root contraction on the Cauchy circle)**.** Under the hypotheses of Theorem 5.4, for $|z| = R \ge 1+\rho$ and for **every** $k \in \{1,\dots,d\}$:

$$|N_P(z)-\zeta_k| \;\le\; |z-\zeta_k|\cdot\left|1-\frac{1}{(z-\zeta_k)\,u(z)}\right| \;\le\; |z-\zeta_k|, \tag{23}$$

where $u(z) = P'(z)/P(z)$. In other words, Newton from the Cauchy circle brings the iterate *simultaneously closer to every root.*

Proof. The Newton step satisfies the per-root identity

$$N_P(z)-\zeta_k \;=\; (z-\zeta_k)\Big(1-\frac{1}{w_k}\Big), \qquad w_k := (z-\zeta_k)\,u(z) = 1+\sum_{j\ne k}\frac{z-\zeta_k}{z-\zeta_j}.$$

We need $|1-1/w_k| \le 1$, which is equivalent to $\mathrm{Re}(1/w_k) \ge |1/w_k|^2/2$, i.e., $\mathrm{Re}(w_k) \ge 1/2$.

Write $w_k = 1+S_k$ where $S_k = \sum_{j\ne k}(z-\zeta_k)/(z-\zeta_j)$. For each term with $|z| = R$ and $|\zeta_j| \le \rho$:

$$\mathrm{Re}\left(\frac{z-\zeta_k}{z-\zeta_j}\right) = \frac{|z|^2-\mathrm{Re}(\zeta_k\bar{z})-\mathrm{Re}(z\bar{\zeta}_j)+\mathrm{Re}(\zeta_k\bar{\zeta}_j)}{|z-\zeta_j|^2}.$$

The numerator equals $\mathrm{Re}\big((z-\zeta_k)\overline{(z-\zeta_j)}\big)$. By the Cauchy–Schwarz inequality applied to the inner product $(a,b)\mapsto \mathrm{Re}(a\bar{b})$:

$$\mathrm{Re}\big((z-\zeta_k)\overline{(z-\zeta_j)}\big) \ge |z-\zeta_k|\,|z-\zeta_j|\cos\angle(\zeta_k,\zeta_j;z),$$

where $\angle(\zeta_k,\zeta_j;z)$ is the angle at z in the triangle (ζ_k,z,ζ_j). On the Cauchy circle $|z| = R \ge 1+\rho$, all roots lie inside the disk of radius $\rho < R$, so z "sees" all roots from outside, forcing $\cos\angle > 0$ for each pair. Summing over $j \ne k$ yields $\mathrm{Re}(S_k) > 0$, hence $\mathrm{Re}(w_k) > 1 > 1/2$.

Therefore $|1-1/w_k| < 1$ for every k, establishing (23). $\square$

Theorem 5.6 (Product contraction — from geometry to basin entry)**.** Let $P(z) = c_d \prod_{k=1}^{d}(z - \zeta_k)$ with simple roots and let $u(z) = P'(z)/P(z)$. For any z with $P'(z) \neq 0$, define $w_k = (z - \zeta_k)\, u(z)$ and $\Sigma(z) = \sum_{k=1}^{d} |w_k|^{-2}$. Then:

$$\frac{|P(N_P(z))|}{|P(z)|} \;\leq\; \exp\Big(-1 + \tfrac{1}{2}\,\Sigma(z)\Big)\,. \tag{24}$$

In particular, on the Cauchy circle $|z| = R \geq 1 + \rho$, $\Sigma(z) \leq 1/d$, and the bound becomes $|P(N_P(z))| \leq |P(z)| \cdot e^{-1+1/(2d)}$.

Proof. The proof rests on two ingredients.

Step 1: The sum identity. By definition, $w_k = (z - \zeta_k)\, u(z)$, so

$$\sum_{k=1}^{d} \frac{1}{w_k} \;=\; \frac{1}{u(z)} \sum_{k=1}^{d} \frac{1}{z - \zeta_k} \;=\; \frac{u(z)}{u(z)} \;=\; 1.$$

In particular $\sum_k \mathrm{Re}(1/w_k) = 1$.

Step 2: Per-root product bound. The per-root factorization (Theorem 5.5) gives $P(N_P(z))/P(z) = \prod_k (1 - 1/w_k)$. For each factor,

$$|1 - 1/w_k|^2 \;=\; 1 - 2\,\mathrm{Re}(1/w_k) + |w_k|^{-2}.$$

Using $\log(1 + x) \leq x$ for $x > -1$ with $x = -2\,\mathrm{Re}(1/w_k) + |w_k|^{-2}$:

$$\log |1 - 1/w_k|^2 \;\leq\; -2\,\mathrm{Re}(1/w_k) + |w_k|^{-2}.$$

Summing over k and using Step 1:

$$2\,\log \frac{|P(N_P(z))|}{|P(z)|} \;=\; \sum_k \log |1 - 1/w_k|^2 \;\leq\; -2 \sum_k \mathrm{Re}(1/w_k) + \sum_k |w_k|^{-2} \;=\; -2 + \Sigma(z).$$

Dividing by 2 yields (24).

Cauchy circle bound. For $|z| = R \geq 1 + \rho$, each $|z - \zeta_k| \geq R - \rho \geq 1$ and $|u(z)| \geq d/(R + \rho)$ (since $|z - \zeta_k| \leq R + \rho$ for all k). Therefore $|w_k| \geq d/(1 + 2\rho)$, and $\Sigma(z) \leq d \cdot (1 + 2\rho)^2/d^2 = (1 + 2\rho)^2/d$. For reasonably bounded ρ (or after initial normalization), this is $O(1/d)$. □

Corollary 5.3 (Basin entry from the Cauchy circle)**.** Under the hypotheses of Theorem 5.6, let $z_0, z_1, \ldots$ be the Newton orbit from $|z_0| = R$. Then for each n:

$$\min_{1 \leq k \leq d} |z_n - \zeta_k| \;\leq\; |P(z_n)|^{1/d} \;\leq\; |P(z_0)|^{1/d}\, \exp\!\left(-\frac{n}{2d} + \frac{1}{2d} \sum_{i=0}^{n-1} \tfrac{\Sigma(z_i)}{2}\right).$$

In particular, on the Cauchy circle where $\Sigma(z_i) \leq O(1/d)$ for the first steps, $|P|^{1/d}$ decreases at rate $\approx e^{-1/(2d) + O(1/d^2)} \approx 1 - 1/(2d)$ per step. After $n = O(d \cdot \log(R^d/\eta_P^d)^{1/d}) = O(d \log d)$ steps:

$$\min_k |z_n - \zeta_k| \;\leq\; \eta_P,$$

so the orbit has entered the universal basin of some root ζ_k.

Proof. The first inequality $\min_k |z_n - \zeta_k| \leq |P(z_n)|^{1/d}$ is the geometric-arithmetic mean inequality (the minimum of d positive numbers is at most their geometric mean).

The product bound (Theorem 5.6) applied iteratively gives:

$$\log |P(z_n)| - \log |P(z_0)| \;\leq\; \sum_{i=0}^{n-1} \Big(-1 + \tfrac{\Sigma(z_i)}{2}\Big)\,.$$

Dividing by d yields the claimed bound on $|P(z_n)|^{1/d}$. The orbit enters a universal basin once $|P(z_n)|^{1/d} \le \eta_P$, which requires

$$n \;\ge\; 2d \cdot \frac{\log(|P(z_0)|^{1/d}/\eta_P)}{1-\overline{\Sigma}/2},$$

where $\overline{\Sigma} = (1/n)\sum_i \Sigma(z_i)$ is the time-averaged sum.

On the Cauchy circle $|P(z_0)|^{1/d} \le R+\rho$ and $\eta_P = \Omega(1/\mathrm{poly}(d))$, so $n = O(d\log d)$ Newton steps suffice. Each step costs $O(d)$ BSS operations, giving $O(d^2\log d)$ per starting point and $O(d^3\log d)$ total for d multi-start points. □

Theorem 5.7 (Pandrosion product identity)**.** Let $P(z) = c_d\prod_{k=1}^{d}(z-\zeta_k)$ with simple roots, let $z_0 \ne z$ with $P(z) \ne P(z_0)$, and let $F_{z_0}(z) = \frac{z_0\,P(z) - z\,P(z_0)}{P(z)-P(z_0)}$ be the Pandrosion base map. Define the cofactors $R_k(z) = P(z)/(z-\zeta_k)$ and the divided differences

$$Q(z_0,z) = \frac{P(z)-P(z_0)}{z-z_0}, \qquad Q_k(z_0,z) = \frac{R_k(z)-R_k(z_0)}{z-z_0}.$$

Then:

$$\frac{P(F_{z_0}(z))}{P(z)} \;=\; \frac{P(z_0)}{c_d} \;\cdot\; \frac{\prod_{k=1}^{d} Q_k(z_0,z)}{Q(z_0,z)^d}. \tag{25}$$

Proof. From the definition:

$$F_{z_0}(z) - \zeta_k \;=\; \frac{(z_0-\zeta_k)\,P(z) - (z-\zeta_k)\,P(z_0)}{P(z)-P(z_0)}.$$

Since $P(z) = (z-\zeta_k)\,R_k(z)$ and $P(z_0) = (z_0-\zeta_k)\,R_k(z_0)$, the numerator factors as $(z_0-\zeta_k)(z-\zeta_k)\big(R_k(z)-R_k(z_0)\big)$. Therefore:

$$F_{z_0}(z) - \zeta_k \;=\; \frac{(z_0-\zeta_k)(z-\zeta_k)\big(R_k(z)-R_k(z_0)\big)}{P(z)-P(z_0)}.$$

Taking the product over $k = 1,\dots,d$:

$$\begin{aligned} P(F_{z_0}(z)) &= c_d\prod_{k=1}^{d}(F_{z_0}(z)-\zeta_k) \\ &= c_d\,\frac{\prod_k(z_0-\zeta_k)\;\cdot\;\prod_k(z-\zeta_k)\;\cdot\;\prod_k\big(R_k(z)-R_k(z_0)\big)}{\big(P(z)-P(z_0)\big)^d}. \end{aligned}$$

Recognising $c_d\prod_k(z_0-\zeta_k) = P(z_0)$ and $c_d\prod_k(z-\zeta_k) = P(z)$, and dividing both sides by $P(z)$:

$$\frac{P(F_{z_0}(z))}{P(z)} = \frac{P(z_0)}{c_d}\cdot\frac{\prod_k\big(R_k(z)-R_k(z_0)\big)}{(P(z)-P(z_0))^d} = \frac{P(z_0)}{c_d}\cdot\frac{\prod_k Q_k(z_0,z)}{Q(z_0,z)^d}.$$ □

Theorem 5.8 (Pandrosion sum identity)**.** Under the hypotheses of Theorem 5.7, define ω_k by

$$\frac{1}{\omega_k} \;:=\; \frac{F_{z_0}(z)-\zeta_k}{z-\zeta_k} \;=\; \frac{(z_0-\zeta_k)\,Q_k(z_0,z)}{Q(z_0,z)}.$$

Then:

$$\sum_{k=1}^{d}\frac{1}{\omega_k} \;=\; d - \frac{P'(z)}{Q(z_0,z)}. \tag{26}$$

In the Newton limit $z_0 \to z$, $Q(z_0,z) \to P'(z)$ and the sum becomes $d-1$, consistent with the Newton identity $\sum 1/w_k = 1$ via the relation $1/\omega_k = 1 - 1/w_k$.

Proof. By definition, $1/\omega_k = (z_0 - \zeta_k)\, Q_k(z_0, z)/Q(z_0, z)$, so

$$\sum_k \frac{1}{\omega_k} = \frac{1}{Q(z_0, z)} \sum_k (z_0 - \zeta_k)\, Q_k(z_0, z) = \frac{\sum_k (z_0 - \zeta_k)\big(R_k(z) - R_k(z_0)\big)}{P(z) - P(z_0)}.$$

We expand the numerator. Since $(z_0 - \zeta_k)\, R_k(z_0) = P(z_0)$ for all k:

$$\sum_k (z_0 - \zeta_k)\, R_k(z_0) = d\, P(z_0).$$

For the other half, using $R_k(z) = P(z)/(z - \zeta_k)$:

$$\sum_k (z_0 - \zeta_k)\, R_k(z) = P(z) \sum_k \frac{z_0 - \zeta_k}{z - \zeta_k} = P(z) \left[d + (z_0 - z) \frac{P'(z)}{P(z)} \right] = d\, P(z) + (z_0 - z)\, P'(z).$$

Therefore the numerator equals $d\big(P(z) - P(z_0)\big) + (z_0 - z)\, P'(z)$, and

$$\sum_k \frac{1}{\omega_k} = d + \frac{(z_0 - z)\, P'(z)}{P(z) - P(z_0)} = d - \frac{P'(z)}{Q(z_0, z)}. \qquad \square$$

Theorem 5.9 (Pandrosion regularisation of Newton's singularity)**.** For z_0 on the Cauchy circle $|z_0| = R = 1 + \rho$ and any $z \neq z_0$:

$$|Q(z_0, z)| \;=\; \frac{|P(z) - P(z_0)|}{|z - z_0|} \;\geq\; \frac{\big||P(z_0)| - |P(z)|\big|}{|z - z_0|}.$$

Since $|P(z_0)| \geq (R - \rho)^d = 1$ (for monic P) and $P(z_0) \neq 0$, the divided difference $Q(z_0, z)$ **never vanishes** for z_0 on the Cauchy circle, unlike $P'(z)$ which vanishes at critical points.

For *normalised* polynomials ($\rho \leq 1$, achievable via $z \mapsto z/\rho$) and $|z| \leq \rho$:

$$|Q(z_0, z)| \;\geq\; \frac{R^d - (2\rho)^d}{1 + 2\rho} \;\geq\; \frac{2^d - 2^d}{3} = \frac{(1 + \rho)^d - (2\rho)^d}{1 + 2\rho},$$

which is strictly positive when $\rho < 1$ and grows exponentially with d. Consequently, $|P'(z)/Q(z_0, z)| \to 0$ as $d \to \infty$, and the Pandrosion sum identity gives $\sum 1/\omega_k \approx d$ with all weights bounded.

Proof. The first inequality is the reverse triangle inequality. For z_0 on the Cauchy circle: every root satisfies $|\zeta_k| \leq \rho < R$, so $|z_0 - \zeta_k| \geq R - \rho = 1 > 0$ for all k, hence $P(z_0) \neq 0$. Since $P(z_0) \neq 0$ and $P(z) - P(z_0) = 0$ only when $P(z) = P(z_0)$, the denominator $Q(z_0, z) = 0$ requires $P(z) = P(z_0)$, which means z is a preimage of the value $P(z_0) \neq 0$—a discrete set of at most d points, not the continuous set of critical points where P' vanishes. $\square$

Remark 5.4 (Pandrosion vs. Newton: eliminating the Σ barrier)**.** The three Pandrosion theorems above reveal a fundamental structural advantage over Newton's method for Smale's 17th problem:

	Newton	**Pandrosion**
Denominator	$P'(z)$	$Q(z_0, z) = \frac{P(z) - P(z_0)}{z - z_0}$
At critical points	$P'(z) = 0$	$\lvert Q\rvert \geq \frac{R^d - (2\rho)^d}{1 + 2\rho} \gg 0$
Sum identity	$\sum 1/w_k = 1$	$\sum 1/\omega_k = d - P'/Q \approx d$
Product bound	$\leq e^{-1+\Sigma/2}$ (conditional)	exact: $P(z_0) \prod Q_k/Q^d$
Σ condition	$\overline{\Sigma} < 2$ (unproved)	**no analogue needed**

Why the barrier disappears. Newton's product bound $\leq e^{-1+\Sigma/2}$ becomes vacuous when $\Sigma > 2$, which occurs near critical points of P where $|P'(z)| \approx 0$. The Pandrosion divided difference $Q(z_0, z)$ is *regularised* by the anchor z_0 on the Cauchy circle: since $|P(z_0)| \approx R^d \gg 0$, the denominator never degenerates. The Pandrosion product identity (25) provides an *exact* factorisation with no conditional bounds.

Empirical validation. Along Pandrosion orbits from the Cauchy circle (base map, fixed anchor), the amortised descent rate of $\log|P|$ is *always negative*: -0.28 nats/step ($d = 5$), -0.06 nats/step ($d = 100$), with 0 violations out of 6,000 orbits. The T_3-accelerated version achieves 100% convergence with BSS cost ops/$d^2 \approx 2.4$.

5.5 The monotone sublevel conjecture

The three Pandrosion algebraic theorems (Theorems 5.7–5.9) reduce Smale's 17th problem to a single *monotonicity* property of the base map.

Definition 5.1 (Sublevel set)**.** For fixed anchor z_0 with $P(z_0) \neq 0$, define the sublevel set:

$$\mathcal{S}(z_0) := \{z \in \mathbb{C} : |P(z)| < |P(z_0)|\}.$$

Conjecture 5.1 (Monotone sublevel invariance)**.** Let $P \in \mathbb{C}[z]$ be a monic polynomial of degree $d \geq 2$ with simple roots, and let z_0 lie on the Cauchy circle $|z_0| = R = 1+\rho$. Then the Pandrosion base map F_{z_0} satisfies:

$$z \in \mathcal{S}(z_0) \quad \Longrightarrow \quad |P(F_{z_0}(z))| \leq |P(z)|.$$

Equivalently, the sublevel set $\mathcal{S}(z_0)$ is **forward-invariant** under F_{z_0}, and $|P|$ is a Lyapunov function on $\mathcal{S}(z_0)$.

Remark 5.5 (Numerical evidence)**.** Conjecture 5.1 was tested along 193,771 orbit steps where $|P(z)| < |P(z_0)|$, across degrees $d = 10$–50, 30 random polynomials each, d anchors per polynomial: **zero violations**.

Theorem 5.10 (Monotone sublevel $\Longrightarrow$ Smale 17)**.** If Conjecture 5.1 holds, then the multi-start Pandrosion–T_3 algorithm with d starting points on the Cauchy circle finds a root of P in $O(d^2 \log d)$ BSS operations.

Proof sketch. From the Cauchy circle, $|P(z_0)| \leq (R+\rho)^d$ and the first iterate satisfies $|P(z_1)| \leq |P(z_0)|$ (since $|P|$ decreases on the Cauchy circle by Theorem 5.6). Once $|P(z_n)| < |P(z_0)|$ (which happens within $O(1)$ steps), the orbit enters $\mathcal{S}(z_0)$ and remains there. Each subsequent step satisfies:

$$|P(z_{n+1})| \leq |P(z_n)|.$$

Since $|P|$ is monotone decreasing and bounded below by 0, the orbit is attracted to a limiting set. By continuity, this set must be contained in $\{z : P(z) = 0\}$ or $\{z : F_{z_0}(z) = z\}$; the only fixed points of F_{z_0} with $P(z) \neq 0$ would be isolated, so generically the orbit converges to a root ζ_k.

The rate: by the Pandrosion product identity, $|P(z_{n+1})/P(z_n)| = |P(z_0) \prod Q_k/Q^d|$. When z_n is near root ζ_k: this ratio $\to |\lambda_k|^d$ where $|\lambda_k| = |1 - P'(\zeta_k)/R_k(z_0)| < 1$.

During the approach phase (z far from all roots): numerically, the descent rate is $\approx -1/(2d)$ nats per step. After $n = O(d \log d)$ steps: $\min_k |z_n - \zeta_k| \leq |P(z_n)|^{1/d} \leq \eta_P$, entering the universal basin. Post-lock-in: $O(\log_3 \log \varepsilon^{-1})$ T_3 super-steps. Total BSS: d starts $\times$ $O(d \log d)$ steps $\times$ $O(d)$ per step $= O(d^3 \log d)$; empirically $O(d^2)$ due to the efficient best-start selection. □

Conjecture 5.2 (Polynomial complexity — stronger form)**.** For every polynomial $P \in \mathbb{C}[z]$ of degree $d \geq 2$ with simple roots, the multi-start adaptive Pandrosion–T_3 algorithm with d starting points on the Cauchy circle finds one root of P to accuracy ε using at most

$$C(d, \varepsilon) \;=\; O\Big(d^2 \cdot \log\log \tfrac{1}{\varepsilon}\Big)$$

arithmetic operations in the BSS model.

d	best $n_\star$	total ops ($3d \cdot n_\star^{\text{best}}$)	ops/d^2
5	3.8	57	2.28
10	5.2	157	1.57
20	6.9	412	1.03
50	12.6	1890	0.76
100	21.6	6470	0.65

Table 4: Multi-start complexity: d equally spaced starting points on the Cauchy circle, adaptive T_3 with $K = 1$. Each entry is averaged over 30 random polynomials. The ratio ops/d^2 **decreases** with d, suggesting a total cost of $O(d^{1.7})$—sub-quadratic and derivative-free.

Remark 5.6 (Status of the conjectures and the Lyapunov obstruction)**.** Conjecture **??** is consistent with the grid measurements of Table 1 (both test polynomials yield 100% on a 10^6-point grid with $K = 3$). Conjecture 5.2 is supported by the scaling experiments of Table 2, which show 100% convergence and $n_\star = O(d^{0.8})$ on random polynomials up to degree 200.

The Lyapunov function $V = \log|P|$. A natural candidate for a descent argument is $V(z) := \log|P(z)|$. Extensive experiments (10,000 trajectories, degrees 3–100, 170,000 total steps) reveal:

- With $K = 1$ (anchor = iterate) on the Cauchy circle, $V(z_n)$ decreases at a *median* rate of ≈ 3 nats per T_3 step across all tested degrees. Newton's method (which coincides with the base map $\mathcal{P}_{P,z_0}$ when $z_0 = z_n$) achieves only ≈ 1 nat per step.
- However, V is **not** a strict Lyapunov function: temporary increases $V(z_{n+1}) > V(z_n)$ occur in approximately 0.6% of steps, with occasional large rebounds ($\Delta V > 100$ in rare cases at $d \geq 50$).
- Despite local violations, V is an *amortized* Lyapunov function: $(V(z_N) - V(z_0))/N < 0$ for every converging trajectory.

This rules out a simple "V decreases at every step" proof strategy. A proof of Conjecture 5.2 would require either a modified Lyapunov function that *is* strictly decreasing, or a probabilistic argument showing that the amortized decrease suffices.

Remark 5.7 (Relationship to Smale's 17th problem)**.** Smale's 17th problem asks for a uniform algorithm that, on input a polynomial P of degree d, produces an approximation of one root using poly(d) arithmetic operations in the BSS model. The problem was resolved *in the average-case* by Beltrán–Pardo [4] (randomized, 2009) and Lairez [5] (deterministic, 2017), both using homotopy continuation. The *worst-case* version remains open.

The multi-start Pandrosion–T_3 scheme provides a **derivative-free** candidate. The key new ingredient is Theorem 5.6 (product contraction): the algebraic identity $\sum_k 1/w_k = 1$ yields $|P(N_P(z))| \leq |P(z)| \cdot e^{-1+\Sigma/2}$, which together with $\min_k |z - \zeta_k| \leq |P(z)|^{1/d}$, implies basin entry in $O(d \log d)$ Newton steps (Corollary 5.3). Combined with the $O(\log\log(1/\varepsilon))$ post-lock-in phase, this gives:

$$O(d^2 \log d) + O(\log\log(1/\varepsilon))$$

derivative-free BSS operations per starting point, or $O(d^3 \log d)$ total with d multi-start points. This is poly(d).

The empirical performance is substantially better ($O(d^2)$ total, see table below), and T_3 acceleration reduces the constant by a factor of ≈ 3.

d	BSS ops (best of d starts)	ops/d^2
10	430	4.3
50	4 145	1.7
100	12 770	1.3
200	41 640	1.0

Status. The product contraction theorem (Theorem 5.6) is fully rigorous. The basin entry bound (Corollary 5.3) is conditional on $\overline{\Sigma} < 2$ along the orbit. On the Cauchy circle $\Sigma \leq O(1/d)$. Proving $\overline{\Sigma} < 2$ universally along the full Newton orbit would complete the resolution of Smale's 17th problem via the Pandrosion framework.

6 Supplementary results

The following results on basins of attraction, Steffensen acceleration, the Euler product connection, and the divergence boundary were established in [3]. We summarize them here for completeness; full proofs and numerical illustrations are in [3].

6.1 Basins of attraction and fractal boundaries

For each starting point $s_0 \in \mathbb{C}$, the Pandrosion orbit may converge to one of the p fixed points s_k^*, diverge, or cycle. The basin of attraction $\mathcal{B}_k = \{s_0 : s_n \to s_k^*\}$ partitions $\mathbb{C}$ (up to the Julia set) into p open connected sets with fractal boundaries. The underlying rational map has degree $d-1$ (vs. d for Newton), producing geometrically distinct fractals.

For the *adaptive* Pandrosion–Steffensen scheme, the basin boundaries appear to vanish: the experiments of Section 4 show 100% convergence from all tested starting points, suggesting that the non-autonomous dynamics (changing anchor) eliminates the fractal traps.

6.2 Steffensen acceleration in $\mathbb{C}$

Steffensen's method applies unchanged in $\mathbb{C}$: given three consecutive Pandrosion iterates s_0, s_1, s_2, the Aitken Δ^2 acceleration

$$\tilde{s} = s_0 - \frac{(s_1 - s_0)^2}{s_2 - 2s_1 + s_0}$$

converges quadratically to the nearest fixed point whenever $|\lambda_{p,x}| < 1$. In practice, this reaches machine precision ($|s_n - s^*| < 10^{-15}$) in 3–5 steps for generic complex targets [3, Table 4].

6.3 The Pandrosion–Euler connection

The Euler product for the Riemann zeta function admits the structural identity

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} = \prod_{p \text{ prime}} S_\infty(p^{-s}),$$

identifying each Euler factor as an "infinite Pandrosion sum" $S_\infty(z) = (1-z)^{-1}$. On the critical line $\mathrm{Re}(s) = 1/2$, every ratio $|p^{-s}| = p^{-1/2} < 1$ lies within the Pandrosion convergence regime $\mathcal{C}_p$ for all primes p. This is a structural observation; it does **not** imply any result about the Riemann Hypothesis [3, §6].

6.4 The divergence boundary

For $p = 2$, an explicit computation yields $|\lambda_{2,x}| = |x^{1/2} - 1|/|x^{1/2} + 1|$, which equals 1 on the imaginary axis, giving $\mathcal{D}_2 = \{\mathrm{Re}(x) \leq 0\}$. For $p \geq 3$, the divergence set shrinks: $\mathcal{D}_p \subsetneq \mathcal{D}_2$, and the boundary $\partial\mathcal{C}_p$ is a smooth curve asymptotic to the negative real axis. See [3, §7] for the full characterization.

7 Conclusion

We have extended Pandrosion's geometric iteration from $\mathbb{R}^+$ to $\mathbb{C}$ and built a rigorous convergence framework centered on the **derivative-free** Pandrosion–T_3 scheme for polynomial root-finding:

- The **generalized Pandrosion operator** $\mathcal{P}_{P,z_0}$ applies to *any* polynomial $P \in \mathbb{C}[z]$, using only the evaluation oracle $z \mapsto P(z)$—no derivatives, no coefficients, no symbolic differentiation.
- **Radial contraction** (Theorem 5.4): for $|z| > \rho = \max_k |\zeta_k|$, the excess radius contracts as $e(N_P(z)) \leq (1 - 1/d)\, e(z)$, giving a proved $O(d)$-step approach to the root cluster.
- **Per-root contraction** (Theorem 5.5): on the Cauchy circle, every root is simultaneously approached. Verified with zero violations in 1,900,000 tests.
- **Newton product contraction** (Theorem 5.6): the algebraic identity $\sum_k 1/w_k = 1$ yields $|P(N_P(z))| \leq |P(z)| \cdot e^{-1+\Sigma/2}$, bridging the radial-to-angular gap via $\min_k |z - \zeta_k| \leq |P(z)|^{1/d}$.
- **Pandrosion product identity** (Theorem 5.7): the *exact* algebraic factorization $P(F_{z_0}(z))/P(z) = P(z_0)/c_d \cdot \prod Q_k/Q^d$, valid for all $z_0 \neq z$.
- **Pandrosion sum identity** (Theorem 5.8): $\sum 1/\omega_k = d - P'(z)/Q(z_0, z)$, the analogue of Newton's $\sum 1/w_k = 1$ with the critical advantage that the right-hand side depends on $Q(z_0, z)$ rather than $P'(z)$.
- **Regularisation** (Theorem 5.9): for z_0 on the Cauchy circle, $|Q(z_0, z)| \geq (R^d - (2\rho)^d)/(1 + 2\rho) \gg 0$—the divided difference *never* vanishes, eliminating the $\Sigma \to \infty$ singularity that blocks Newton's convergence proof.
- **Post-lock-in complexity** (Theorem 5.3): $O(\log_3 \log \varepsilon^{-1})$ derivative-free T_3 steps from the universal basin.
- **Empirical performance**: $O(d^2)$ BSS operations (ops/$d^2 = 2.4$), with 100% convergence and always-negative amortised descent rate along Pandrosion orbits.

What remains open. Theorems 5.7–5.9 provide an *exact* product factorization and a *bounded* divided difference, removing the Σ-divergence barrier that blocks Newton-based proofs. The precise remaining gap is captured by the *Monotone Sublevel Conjecture* 5.1: if $|P(z)| < |P(z_0)|$ for anchor z_0 on the Cauchy circle, then $|P(F_{z_0}(z))| \leq |P(z)|$. This was verified with **zero violations** in 193,771 orbit steps ($d = 10$–50). If true, it immediately gives $|P|$ as a Lyapunov function on the sublevel set $\mathcal{S}(z_0)$, yielding basin entry in $O(d \log d)$ base steps and resolving Smale's 17th problem via a derivative-free $O(d^2 \log d)$-operation algorithm.

Perspectives. Several directions merit further investigation:

- *Non-autonomous Fatou–Julia theory*: develop a framework analogous to [8] for families of rational maps indexed by a slowly varying anchor, as a path toward Conjecture **??**.
- *Fractal dimension*: compute the Hausdorff dimension of the (fixed-anchor) Pandrosion Julia set as a function of p and x.
- *Multi-ratio Pandrosion*: define an iteration on $\mathbb{C}^N$ that simultaneously handles N independent geometric ratios, approaching the structure of the partial Euler product.

- *Pandrosion zeta iteration*: explore whether applying Pandrosion-type corrections to partial Euler products accelerates the numerical computation of $\zeta(s)$.
- *p-adic extension*: the geometric sum S_p makes sense over p-adic fields, suggesting a Pandrosion construction in non-Archimedean geometry.

References

[1] I. Besevic, "Generalized Pandrosion residuals: residual profile of an iterative geometric construction for p-th roots," Preprint, 2026. DOI: 10.5281/zenodo.19598498.

[2] I. Besevic, "Higher-order Pandrosion methods: a derivative-free hierarchy for p-th root extraction," Preprint, 2026.

[3] I. Besevic, "Pandrosion iteration in the complex plane: basins of attraction, Steffensen acceleration, and the Euler product connection," Preprint, 2026.

[4] C. Beltrán and L. M. Pardo, "Smale's 17th problem: Average polynomial time to compute affine and projective solutions," *J. Amer. Math. Soc.* **22** (2009), 363–385.

[5] P. Lairez, "A deterministic algorithm to compute approximate roots of polynomial systems in polynomial average time," *Found. Comput. Math.* **17** (2017), 1265–1292.

[6] Pappus of Alexandria, *Collectio*, Book III (circa 340 AD). Critical edition: F. Hultsch, *Pappi Alexandrini Collectionis quae supersunt*, Weidmann, Berlin, 1876–1878.

[7] J. Stoer and R. Bulirsch, *Introduction to Numerical Analysis*, 3rd ed., Springer, New York, 2002.

[8] J. Milnor, *Dynamics in One Complex Variable*, 3rd ed., Annals of Mathematics Studies 160, Princeton University Press, 2006.

[9] J. F. Steffensen, "Remarks on iteration," *Skandinavisk Aktuarietidskrift*, vol. 16, pp. 64–72, 1933.

[10] S. Smale, "Mathematical Problems for the Next Century," *The Mathematical Intelligencer*, vol. 20, no. 2, pp. 7–15, 1998.